

Yeh AI Assistant ka **Sab se Powerful aur Zaheen (Intelligent) Brain** hai. Project ke baqi hissay sirf sawal/jawab kerte hain lkn yeh file AI ko ek insaan ki tarha "Sochne aur Action Lene" (ReAct Framework) ki salahiyat deti hai.

File Ka Maqsad (Purpose)

Agar user ChatGPT ki trah puche "Google se check kro MS Dhoni kis saal paida hua, fir calculator pr 2024 se minus kr k uski current umer batao", toh sada AI yeh ni kar sakta. Yeh script usay "Tools" (hathyar/auzarr) use karnay ke kabil banati hai.

Code Blocks aur Classes ki Tafseel

1. class ToolDefinition / ToolParameter

- **Kya karta hai?:** Tools ki setting aur parameters define kr k unhe standard JSON Formats (jo LLMs samajhtay hain) m convert karta ha (schema generator).

2. Built-in Tools (Classes)

Is mein 10 se zyada kamaal ke functions mojud hain: * **CalculatorTool**: Math k sawalat automatically Python k zariye solve karta hai. * **CodeRunnerTool**: (Dangerous Tool) Yeh AI ko ijazat deta ha k wo physical system pe apna Python code run ke k errors test kr sake. * **FileReaderTool / FileWriterTool**: Computer se files parhna aur text files banana/edit kerna. * **WebSearchTool**: DuckDuckGo k zareya internet se live latest info (news/score) dhoondta ha. * **WebScraperTool**: Kisi website ki link extract kar key uska sara lamba content nikalata hai. * **RAGSearchTool / KnowledgeGraphTool**: Local PDFs aur Nodes mein search marta hai. * **AgentMemoryTool**: AI apni batain aur user ki pasandeedgi Database m save or read (recall) kar sakta hai!

3. class ReActAgent

- **Kya karta hai?:** Yeh main engine (loop) hai.

Logic (Kaise kam karta ha):

1. Pehlay yeh LLM ko Tool list bhejta hai.

2. LLM `Thought` generate krti ha: *"User umer poch rha h. Mujhe web search use krni chaye"*.
3. Phr LLM `Action: web_search` generate karta ha.
4. Code isay pakkarr(parse) kar tool chala leta ha aur LLM ko result (Observation) deta ha.
5. Yeh loop bar bar repeat hota ha jb tak answer (Final Answer) nahi mil jata. Wah!